

EDUCATION

2026–2027 **University of Wisconsin–Madison**

M.S. in Physics: Quantum Computing (MSPQC). Expected May 2027.

Relevant coursework: Quantum Computing Laboratory, Qubit Tune-Up and Programming, Solid State, Graduate Seminar on Quantum Computing.

2022–2026 **University of Wisconsin–Madison**

B.S. in Computer Science and Mathematics; Physics Certificate.

Relevant coursework: Intro to Quantum Computing, Systems Architecture for Quantum Computing, Quantum Mechanics, Algorithms, Real Analysis, Abstract Algebra, Combinatorics, Operating Systems, Linear Optimization, Theory of Programming Languages.

RESEARCH

2025– **Distributed Quantum Computing**, Argonne National Laboratory.

Advisor: Dr. Rajkumar Kettimuthu.

- Designed and built **SeQUeNCe.DQC**, a top-level module turning SeQUeNCe (2026 R&D 100 Winner) into a full-stack DQC simulator: circuit front end, compiler, central controller, and per-node worker agents.
- Implemented DQC nodes with separate data and communication memory, TeleGate/TeleData teleportation primitives.
- Built QAP, random, and time-sliced FGP-rOEE compilers behind a compiler-agnostic interface, plus ket-vector trajectory and density-matrix CPTP simulation backends.
- Modeled the classical controller as a real network node, and added a trajectory noise model with per-qubit idle T_1/T_2 decoherence and per-shot Pauli errors.
- Reproduced the 71% distributed Grover success rate of a *Nature* trapped-ion experiment under both lumped and full per-source error budgets.

2024– **Fault Tolerance & Benchmarking**, University of Wisconsin–Madison.

Advisor: Prof. Swamit Tannu.

- Currently developing compilation strategies for bivariate bicycle codes.
- Developed benchmark suites for fault-tolerant quantum hardware; used variational autoencoders to learn latent-space representations of quantum circuits, enabling similarity analysis based on metrics such as T-gate resource estimation.

Apr 2026– **Graph State Verification**, University of Oxford (remote).

Collaborator: Dr. Sathyawageeswar Subramanian.

- Extending graph state verification protocols to more sophisticated Pauli noise models for fault-tolerant measurement-based quantum computing.

2023– **Quantum Algorithms for Scientific Computing**, University of Wisconsin–Madison.

Advisor: Prof. Krishnan Suresh.

- Formulated sparse approximate inverse (SPAI) preconditioner construction as a QUBO and computed it on D-Wave Advantage hardware to accelerate conjugate gradient solves of poorly conditioned linear systems.
- Reduced the box-algorithm contraction ratio for 30–40% faster solution of random

linear systems; adapted quantum annealing to inequality-constrained portfolio optimization.

- Developed efficient encoding schemes for structured quantum states (PyEncode), with applications to engineering simulation.

PUBLICATIONS

- 2027 **SeQUeNCe.DQC: Full-stack Simulation of Distributed Quantum Computing.** Sanjay Suresh, Caitao Zhan, Joaquin Chung, Rajkumar Kettimuthu. *IEEE International Conference on Communications*, under review.
- 2026 **PyEncode: An Open-Source Library for Structured Quantum State Preparation.** Krishnan Suresh, Sanjay Suresh. *Quantum Science & Technology*.
- 2025 **Computing a Sparse Approximate Inverse on Quantum Annealing Machines.** Sanjay Suresh, Krishnan Suresh. *Journal of Computing and Information Science in Engineering*.
- 2024 **Optimal Box Contraction for Solving Linear Systems via Simulated and Quantum Annealing.** Sanjay Suresh, Krishnan Suresh. *Engineering Optimization*.
- 2021 **A Visual Approach to Programming for Middle and High School Students.** Sanjay Suresh. ISBN 979-8470081841.

TALKS & POSTERS

- 2026 Workshop co-instructor, *Quantum Computing for Engineers*, QC 2026; poster on PyEncode.
- 2025 Poster on distributed quantum computing with SeQUeNCe, Chicago Quantum Summit.
- 2024 Talk and poster on quantum annealing for linear systems and constrained optimization, WCCM, Vancouver.

PROJECTS

- 2025–2026 **SurfaceScope**, surface code QEC pipeline and visualizer.
- Compiles OpenQASM circuits to native Clifford gates, routes qubits with SWAP insertion on a linear array, and simulates per-patch error correction with Stim.
 - Estimates thresholds across code distances and physical error rates; visualizes syndrome extraction, detector error models, and MWPM decoding.
- Python, Flask, React, TypeScript, Stim, PyMatching.*
- 2025–2026 **Studia**, AI tutoring platform.
- Full-stack platform with structured lessons, PDF context extraction via RAG, automated review generation, and real-time Manim animations. Deployed in CS 639 (Prof. Swamit Tannu).
- Next.js, TypeScript, Convex, Vercel AI SDK, Manim.*

TEACHING & OUTREACH

- 2019– **Principal Instructor**, Printable Programming, Madison, WI.

- Authored a CAD modelling, programming and 3D-printing curriculum; taught STEM at the Boys and Girls Club of Dane County; contributed to the CADjs software application.

2020–2022 **President & Captain**, Science Bowl, Middleton, WI.

- Founded Summit Scibowl for advanced student training; built online platforms engaging 2,000+ students nationwide and organized national tournaments.

HONORS & AWARDS

- 2022 Atlas Fellowship — selected from 7,000+ applicants worldwide (114 recipients) for a \$50K fellowship.
- 2022 Wisconsin Math Talent Search Finalist; Mathematics Student of the Year, Middleton High School.
- 2021, 2022 Chemistry Olympiad National Qualifier; National Astrophysics Olympiad Qualifier (2021); AIME Qualifier.
- 2020 Naval Research Award, Capital Science and Engineering Fair; Prometheus Science Bowl, 2nd nationally.

SKILLS

- Quantum Qiskit, Stim, PyMatching, PsiQuantum Workbench, SeQUeNCe, pytket / pytket-dqc, PennyLane, QuTiP, D-Wave Ocean.
- Languages Python, TypeScript, C, Java, Haskell, MATLAB.
- Web & Infra Next.js, React, Convex, Vercel AI SDK, Flask, PostgreSQL, Docker, Git.

OTHER

- Music Carnatic violin. Professional violinist and accompanist to frontline artists at major festivals across India and the US. Trained since 2012; performing as soloist and accompanist since 2017.